

CSC11006 - INTRODUCTION TO CLOUD COMPUTING SERVICES

PRACTICE

LAB: Batch Loading Data from Cloud Storage to Big Query

I. General information

ID: LAB
Period: 30 minutes
Deadline:

II. Outcome:

This lab will adapt to these following course's outcomes:

- G2.1
- G4.1

III. Describe:

1. Introduction:

Your team needs to upload objects and data to Cloud Storage while also being able to analyze and load that data into BigQuery. Your task is to set up a batch process to transfer the data from Cloud Storage to BigQuery, enabling your team to progress with its objectives.

2. Prepare:

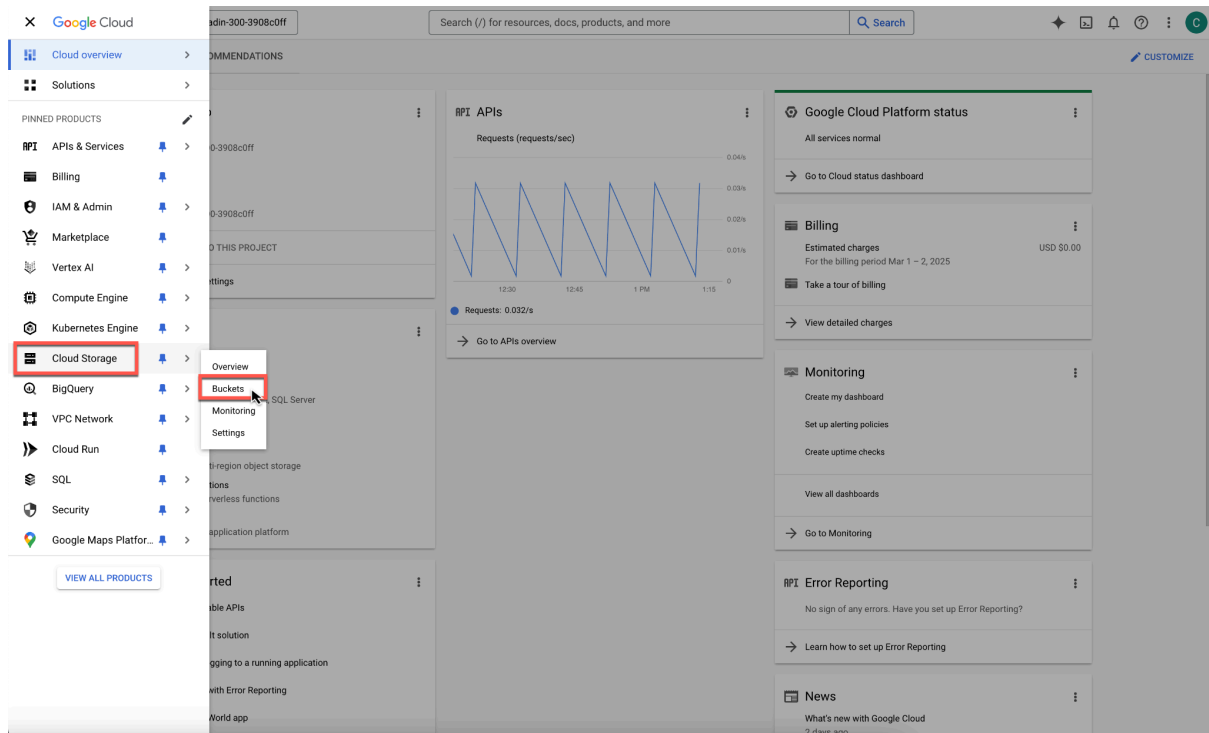
2.1 Download the sneakers1.csv files for this lab from [the course's GitHub repo](#).

2.2 Log In to GCP

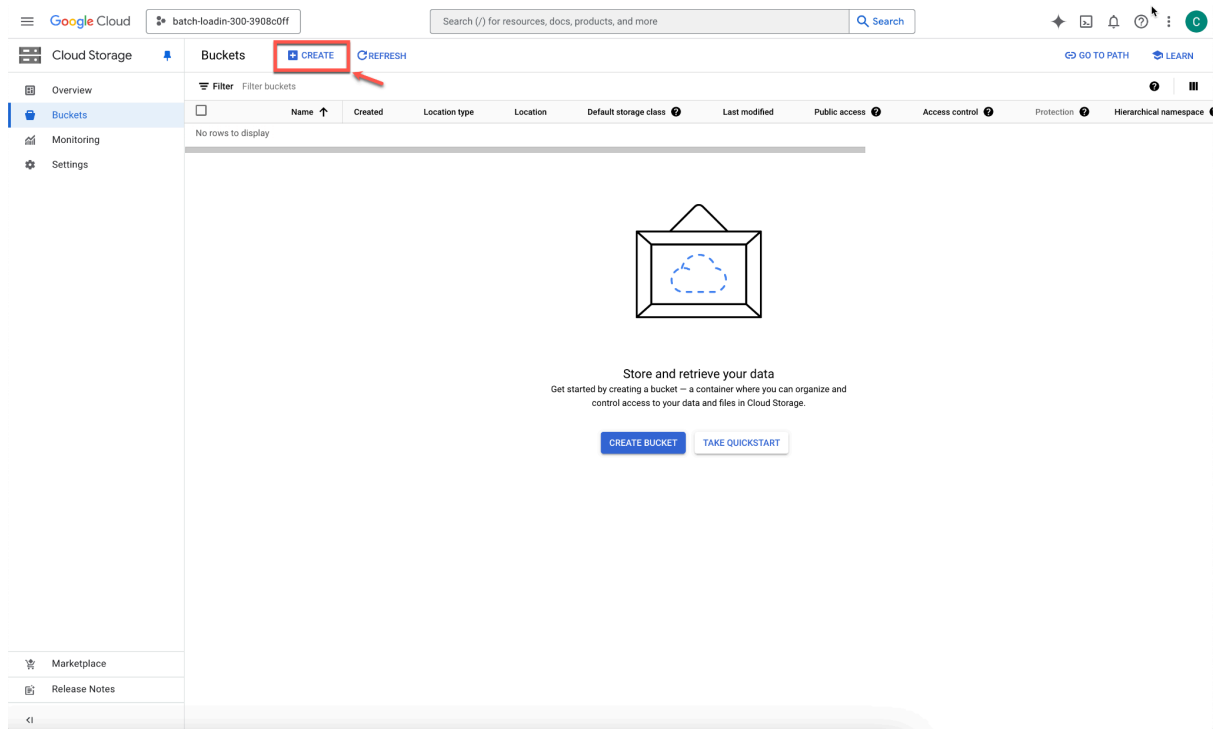
Sign in to Google Cloud Platform using the credentials provided on the lab page.

3. Create a Cloud Storage Bucket

- In the top left corner, click the hamburger menu icon (the icon consisting of three horizontal lines).
- In the left-hand navigation menu, select Cloud Storage.



- Click Create to create a new bucket.



- Under Name your bucket, enter loading-data-bucket followed by some random characters to make the name unique.
- Click Continue.

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Cloud Storage Create a bucket

Overview Buckets Monitoring Settings

Get Started
Pick a globally unique, permanent name. [Naming guidelines](#)
loaddata2025
Tip: Don't include any sensitive information
Optimize storage for data-intensive workloads
Labels (optional)
CONTINUE

Good to know
Location pricing
Storage rates vary depending on the storage class of your data and location of your bucket. [Pricing details](#)
Current configuration: Multi-region / Standard

Item	Cost
us (multiple regions in United States)	\$0.026 per GB-month
With default replication	\$0.020 per GB written

ESTIMATE YOUR MONTHLY COST

Choose where to store your data
Location: us (multiple regions in United States)
Location type: Multi-region

Choose a storage class for your data
Default storage class: Standard

Choose how to control access to objects
Public access prevention: On
Access control: Uniform

Choose how to protect object data
Soft delete policy: Default
Object versioning: Disabled
Bucket retention policy: Disabled
Object retention: Disabled
Encryption type: Google-managed

CREATE CANCEL

Marketplace Release Notes

- Under Location type, keep the default option and click Continue.

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Cloud Storage Create a bucket

Overview Buckets Monitoring Settings

Get Started
Name: loaddata2025

Choose where to store your data
This choice defines the geographic placement of your data and affects cost, performance, and availability. Cannot be changed later. [Learn more](#)
Only locations allowed by the organization's resource location policy can be selected.

Location type
☒ Multi-region
Highest availability across largest area
us (multiple regions in United States)
☐ Add cross-bucket replication via Storage Transfer Service
As data is added or changed, replicate it to another bucket, enabling you to store a copy that follows different bucket settings - e.g., region, storage class, etc. [Learn more](#)
☐ Dual-region
High availability and low latency across 2 regions
☐ Region
Lowest latency within a single region
CONTINUE

Choose a storage class for your data
Default storage class: Standard

Choose how to control access to objects
Public access prevention: On
Access control: Uniform

Choose how to protect object data
Soft delete policy: Default
Object versioning: Disabled
Bucket retention policy: Disabled
Object retention: Disabled
Encryption type: Google-managed

Marketplace Release Notes

- Under Storage class, keep the default and click Continue

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Cloud Storage Overview Buckets Monitoring Settings

Get Started
Name: loaddata2025

Choose where to store your data
Location: us (multiple regions in United States)
Location type: Multi-region

Choose a storage class for your data
A storage class sets costs for storage, retrieval, and operations, with minimal differences in uptime. Choose if you want objects to be managed automatically or specify a default storage class based on how long you plan to store your data and your workload or use case. [Learn more](#)

☐ Autoclass
Automatically transitions each object to Standard or Nearline class based on object-level activity, to optimize for cost and latency. Recommended if usage frequency may be unpredictable. Can be changed to a default class at any time. [Pricing details](#)

☒ **Set a default class**
Applies to all objects in your bucket unless you manually modify the class per object or set object lifecycle rules. Best when your usage is highly predictable.

☒ **Standard**
Best for short-term storage and frequently accessed data

☐ Nearline
Best for backups and data accessed less than once a month

☐ Coldline
Best for disaster recovery and data accessed less than once a quarter

☐ Archive
Best for long-term digital preservation of data accessed less than once a year

CONTINUE

Good to know
Location pricing
Storage rates vary depending on the storage class of your data and location of your bucket. [Pricing details](#)

Current configuration: Multi-region / Standard

Item	Cost
us (multiple regions in United States)	\$0.026 per GB-month
With default replication	\$0.020 per GB written

ESTIMATE YOUR MONTHLY COST

Choose how to control access to objects
Public access prevention: On
Access control: Uniform

Choose how to protect object data
Soft delete policy: Default
Object versioning: Disabled
Bucket retention policy: Disabled
Object retention: Disabled
Encryption type: Google-managed

- Under Access control, keep the default option and click Continue.

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Cloud Storage Overview Buckets Monitoring Settings

Get Started
Name: loaddata2025

Choose where to store your data
Location: us (multiple regions in United States)
Location type: Multi-region

Choose a storage class for your data
Default storage class: Standard

Choose how to control access to objects
Prevent public access
Restrict data from being publicly accessible via the internet. Will prevent this bucket from being used for web hosting. [Learn more](#)

☒ **Enforce public access prevention on this bucket**

Access control

☒ **Uniform**
Ensure uniform access to all objects in the bucket by using only bucket-level permissions (IAM). This option becomes permanent after 90 days. [Learn more](#)

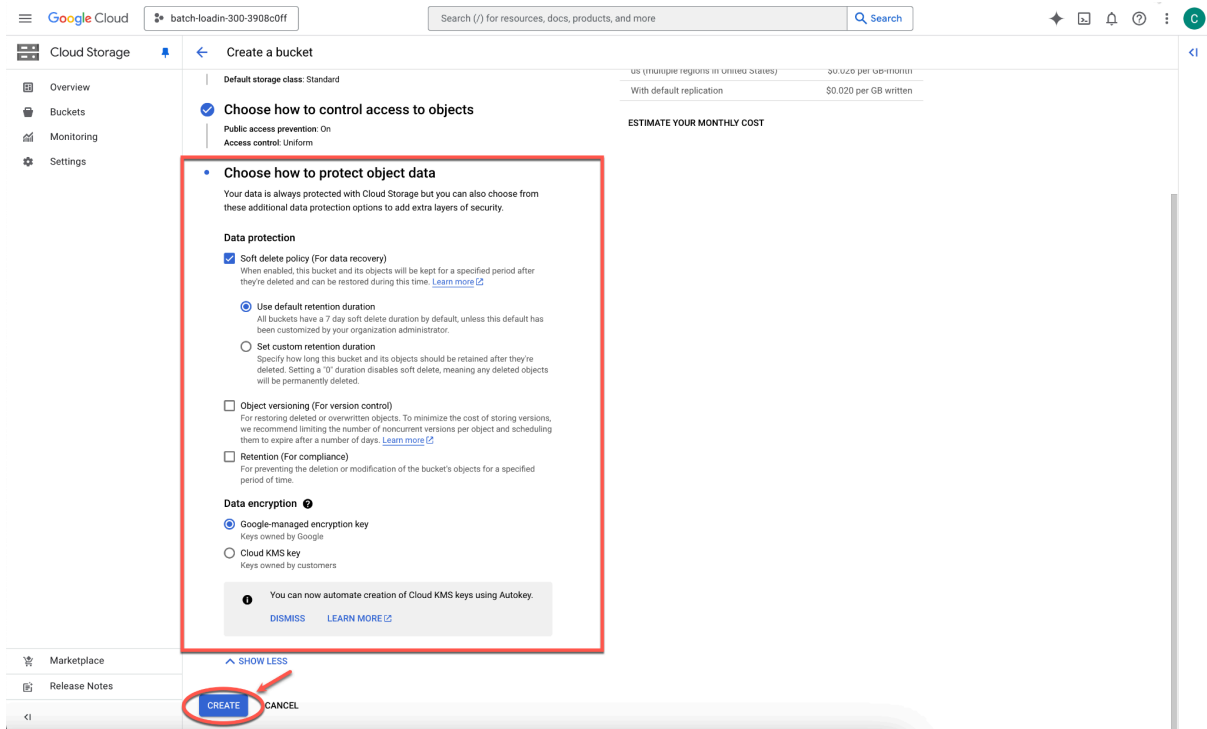
☐ Fine-grained
Specify access to individual objects by using object-level permissions (ACLs) in addition to your bucket-level permissions (IAM). [Learn more](#)

CONTINUE

Choose how to protect object data
Soft delete policy: Default
Object versioning: Disabled
Bucket retention policy: Disabled
Object retention: Disabled
Encryption type: Google-managed

CREATE **CANCEL**

- Under Protection, keep the default option and click Continue.
- Under Data Encryption, select Google-managed encryption key.



Click Create.
Click Confirm.

Public access will be prevented

This bucket is set to prevent exposure of its data on the public internet.

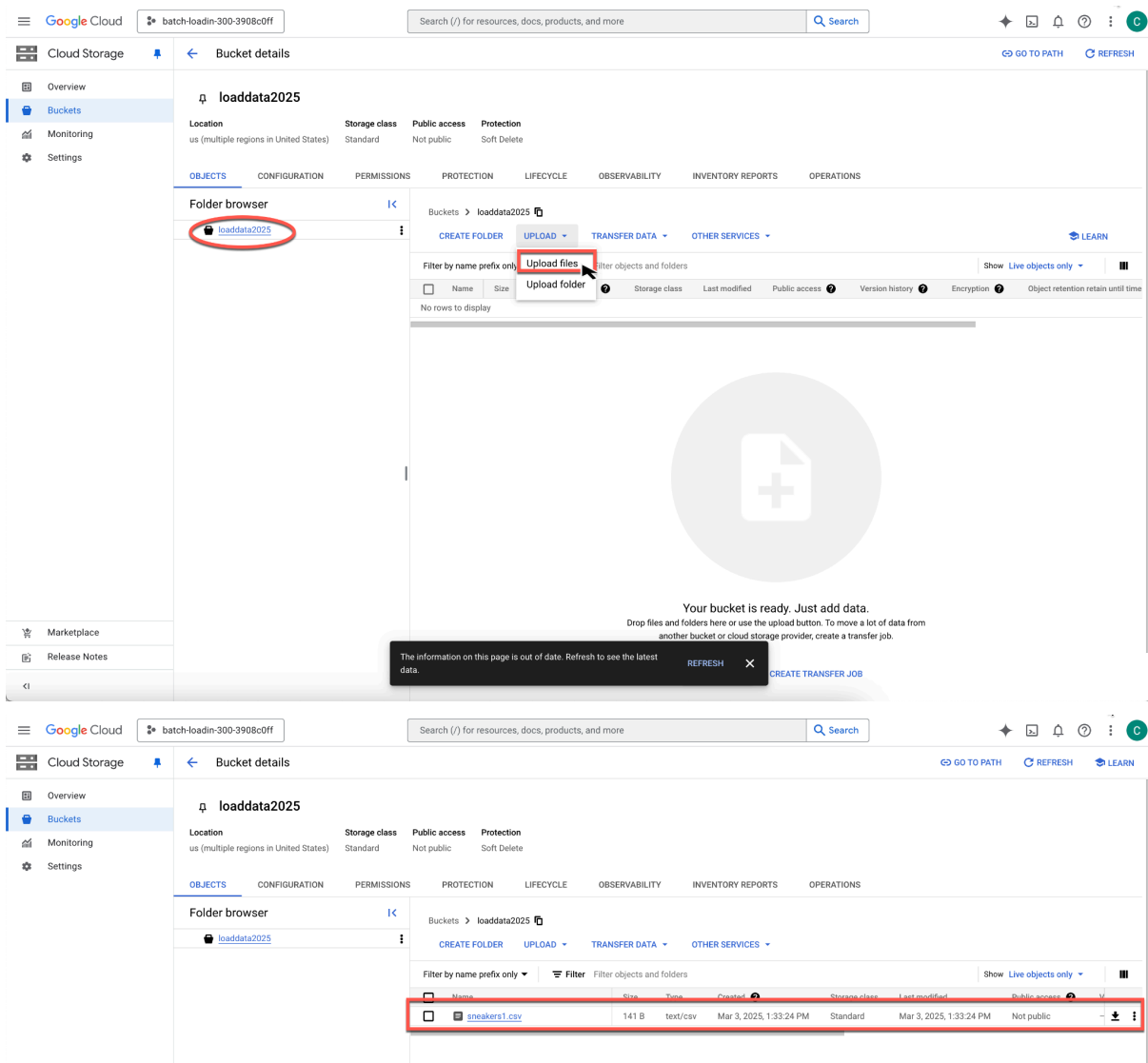
Keep this setting enabled unless you have a use case that requires public access (such as static website hosting). You can change it now or later. [Learn more](#)

- ☒ Enforce public access prevention on this bucket
- ☐ Don't show this message again

CANCEL

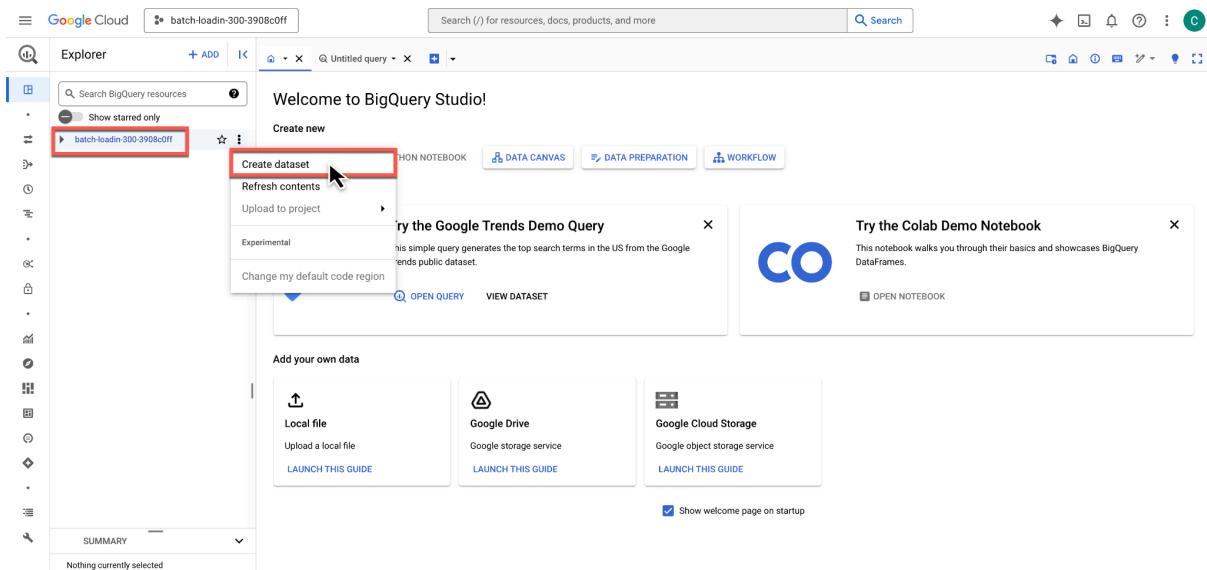
CONFIRM

- Upload a file into the bucket by either dragging a file into the bucket area or clicking Upload and selecting a file to be uploaded.

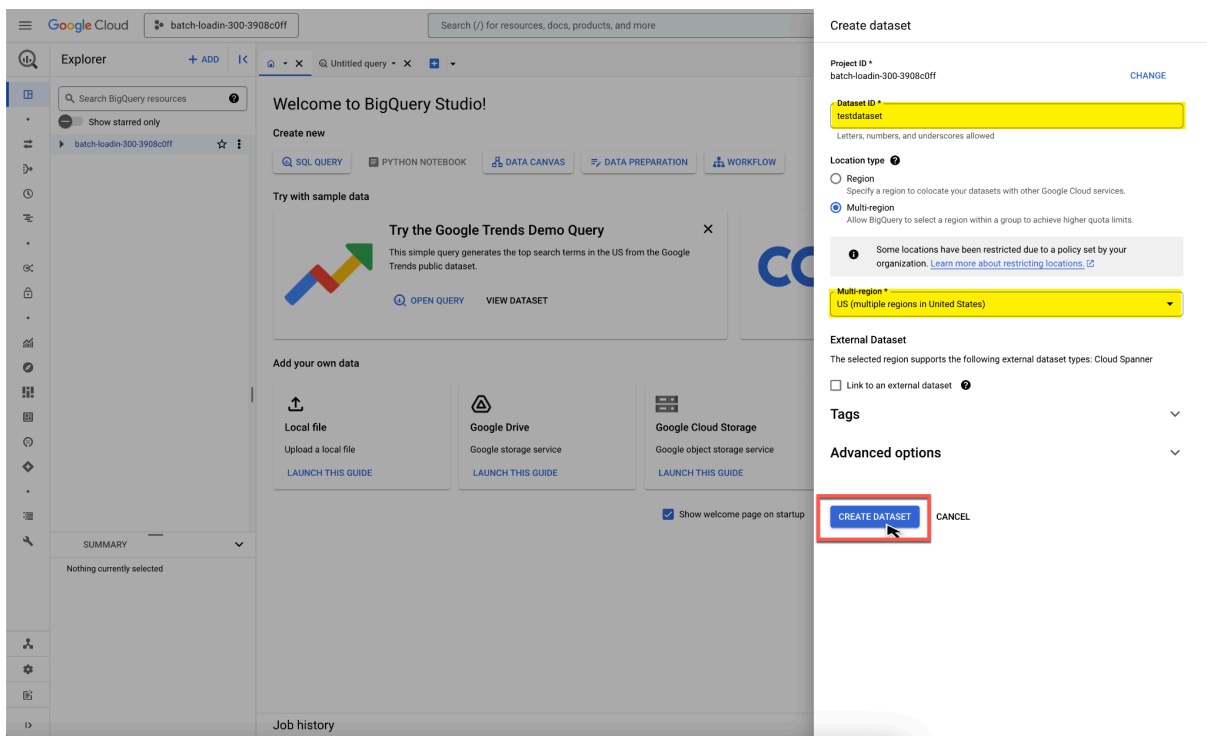


4. Batch Load Data from Cloud Storage to BigQuery

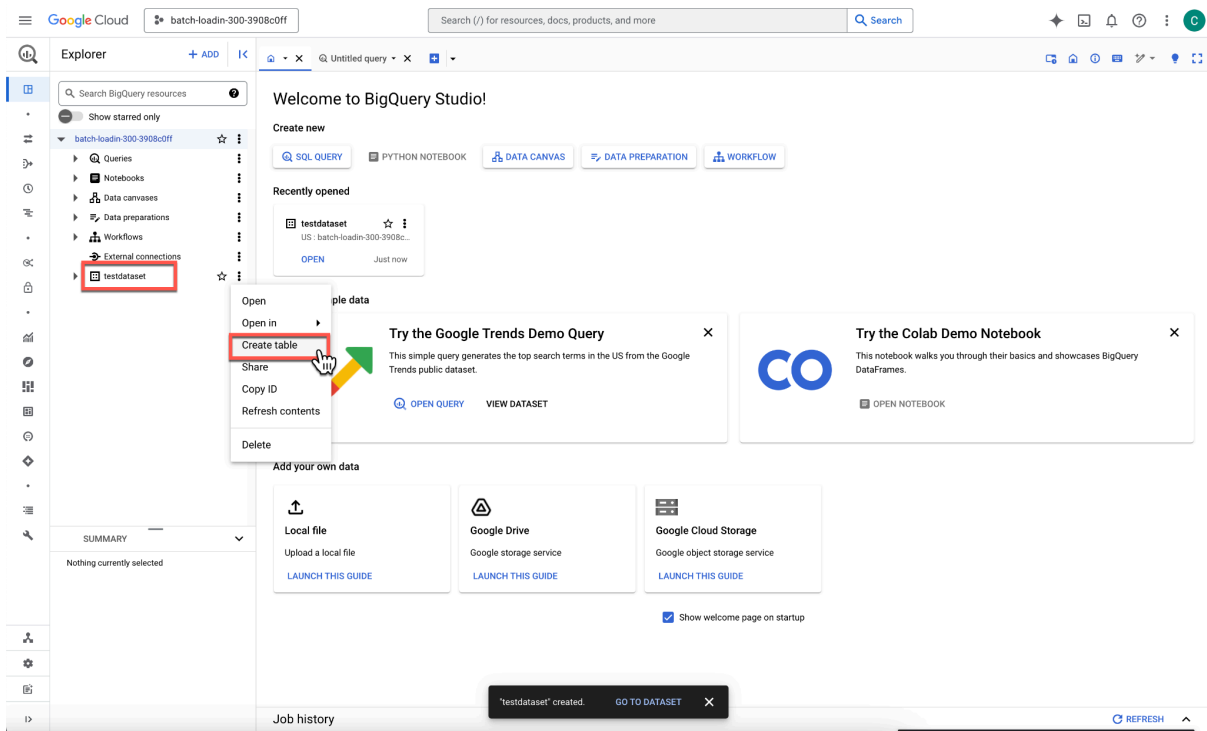
- In the top left corner, click the hamburger menu icon (the icon consisting of three horizontal lines).
- In the left-hand navigation menu, select BigQuery.
- In the popup window, click Done.
- In the left-hand menu, expand the project listed. You should see that we don't have any datasets.
- Click the three-dot icon next to the listed project.
- Click Create dataset.



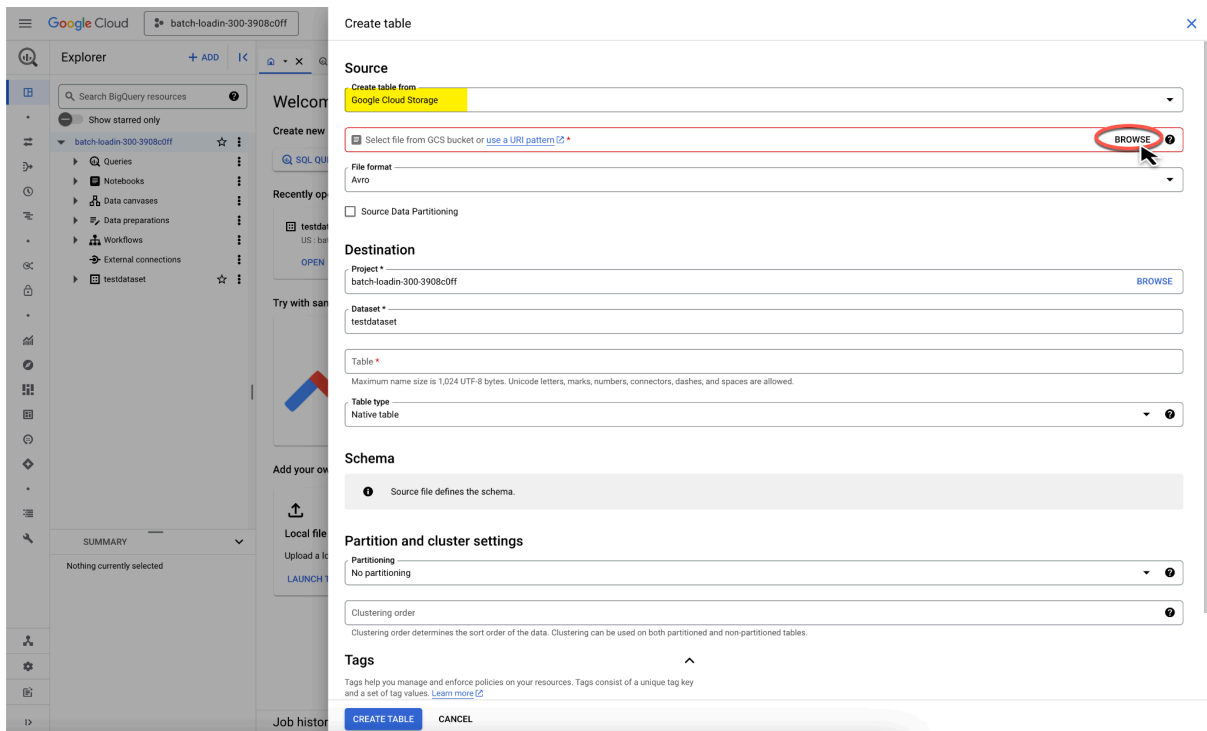
- Under Dataset ID, enter testingdataset.
- Under Data location, select us (multiple regions in United States).
- Click Create Dataset.



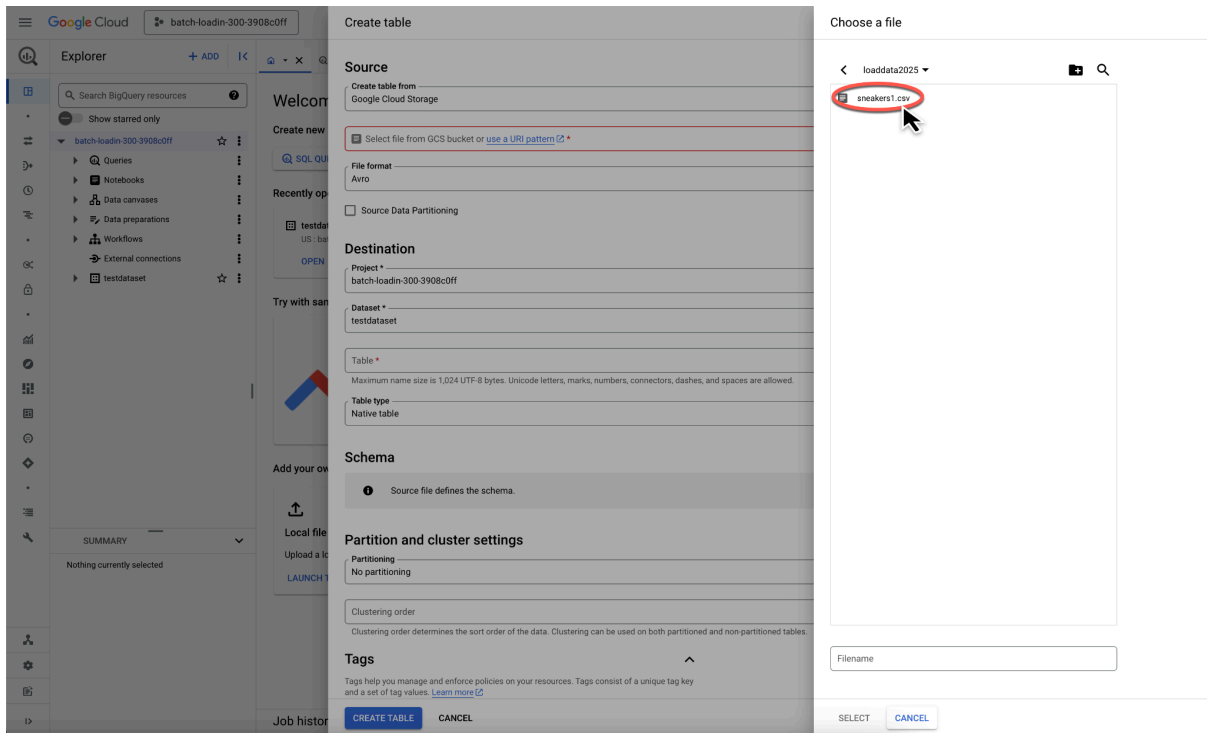
- Next to the newly created testingdataset dataset, click the three-dot icon.
- Click Create table.



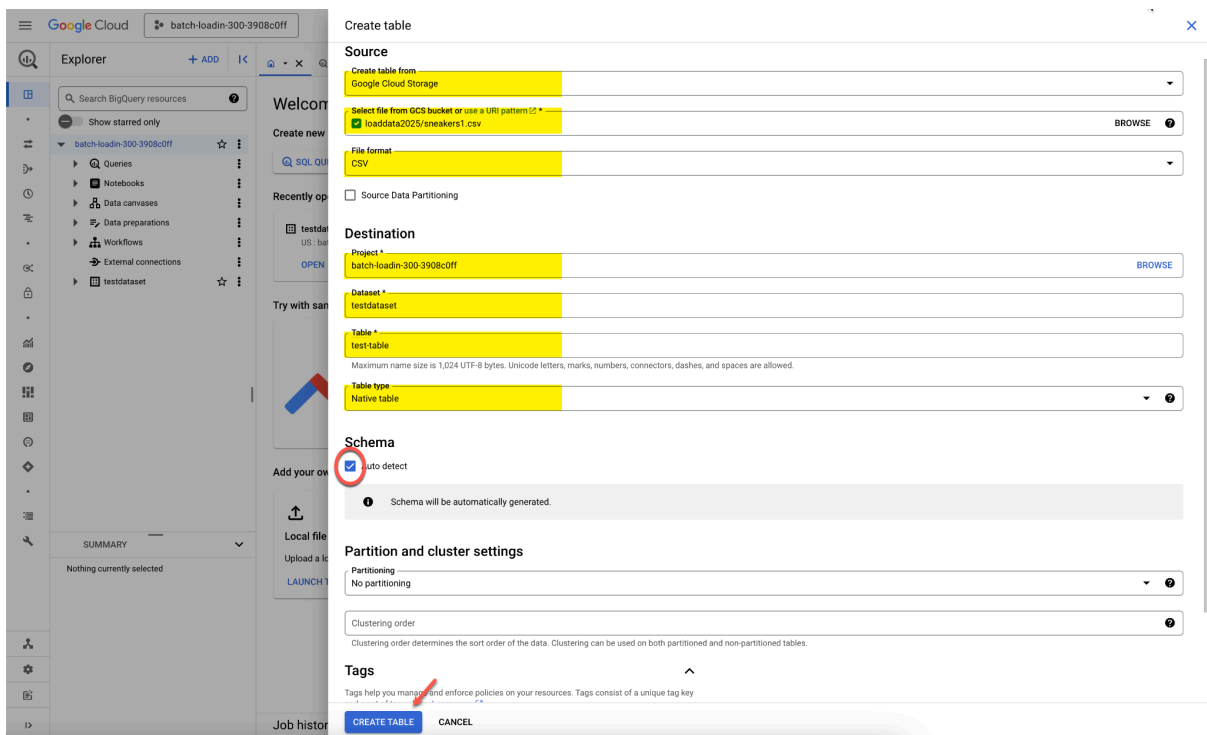
- Under Create table from, select Google Cloud Storage.
- Click Browse.



- Under Buckets, click the bucket that we created.
- Click the file that you uploaded.
- Click Select.



- Under Table, enter test-table.
- Under Schema, select Auto detect.
- Click Create table.



- In the left-hand navigation menu, click test-table.
- Click Open in -> Select In split tab.

The screenshot shows the Google Cloud BigQuery Explorer interface. On the left, the 'test-table' is selected under the 'testdataset'. The main panel displays the table's schema with columns: sneaker_brand (STRING), sneaker_model (STRING), and sneaker_year (INTEGER). The 'OPEN IN' dropdown menu is open, showing various options. The 'PREVIEW' button is highlighted in the top right corner of the table details section.

- In the query pane, enter sneaker_model after SELECT.
- Click Run. You should see query results for sneaker models in the bottom right.
- Click Preview. You should see a table with columns of sneaker_brand, sneaker_model, and sneaker_year.

The screenshot shows the Google Cloud BigQuery Explorer interface. The 'test-table' is selected. The 'Query results' pane is open, showing the results of the query: sneaker_model. The results are displayed in a table with columns: sneaker_brand, sneaker_model, and sneaker_year. The 'Query results' pane is highlighted with a red box.

Row	sneaker_brand	sneaker_model	sneaker_year
1	Adidas	Yeezy 350 Boost	2015
2	Nike	Air Max 90	1990
3	Nike	Air Jordan 1	1985
4	Nike	Air Max 95	1995